# Predicting Early-Stage Parkinson's Disease Using Circadian Keystroke Dynamics and SHAP Clinical Heatmaps

Fayez Ahmad Protik
*BRAC University*

fayez.ahmad.protik@g.bracu.ac.bd

***Abstract*—Traditionally, the diagnosis of Parkinson's Disease (PD) is made on the basis of a clinical examination that reveals motor symptoms. However, clinical evaluations are done only at certain times and do not necessarily reflect the fluctuations that occur from day to day. In this paper, we suggest using machine learning to detect early stage PD from passive, unconstrained keystroke dynamics collected in users' natural home environments. We present a novel clinical feature, 'Circadian HoldTime Fluctuation', which mathematically models the daily 'On/Off' motor fluctuations associated with Levodopa medication cycles by comparing morning and evening typing behaviours. We used the Synthetic Minority Over-sampling Technique (SMOTE) to overcome the significant class imbalance (75% healthy, 25% PD) and we trained and tested several algorithms. The highest robustness was found in a Soft Voting Ensemble that utilized Random Forest, XGBoost and SVM, with an accuracy of 69.2% compared to the baseline of 61.5%. Most importantly, we used the underlying model to generate SHapley Additive exPlanations (SHAP) to meet the tight medical demands for the algorithmic transparency. This Explainable AI (XAI) integration is a visual proof that our model is not based on some random mathematical noise but actual physiological severity indicators like Circadian Fluctuation. The results show that this unconstrained monitoring with understandable AI is a promising non-invasive method for continuous PD tracking.**

***Index Terms*—Parkinson's Disease, Keystroke Dynamics, Machine Learning, Soft Voting Ensemble, Explainable AI (SHAP), Circadian Rhythm, Unconstrained Monitoring.**

## I. Introduction

Parkinson's Disease (PD) is progressive destruction of the dopaminergic neurons that results in symptoms of tremor, bradykinesia, rigidity, and postural instability. The gold standard for diagnosis is based on clinical assessments, including the Unified Parkinson's Disease Rating Scale (UPDRS). However, these assessments are only episodic, subjective and can only be made in a clinic, and are unsuitable for detecting the high frequency motor fluctuations (often referred to as 'On/Off' states) seen in patients when they respond to drug regimens.

Over the past few years, the field of human-computer interaction (HCI) has brought a number of advancements that have pointed to keystroke dynamics as a potential, non-invasive digital biomarker. The timing parameters of typing change significantly over the course of motor degradation, such as the 'HoldTime' (time a key is being held) and 'FlightTime' (time between keys) measures of typing. Most of the literature available uses limited typing tasks, in which the participant is asked to type a predetermined text in an experimental laboratory setting. These controlled studies are successful, but they don't represent actual natural typing behaviors.

This study uses the Tappy Keystroke dataset of more than 9 million keystrokes collected passively from users who interacted with their own personal computers in their normal daily routines for several months. In this paper, we make the following three main contributions to effectively model this highly noisy data:

1) Novel Feature Engineering: We created a novel feature, 'Circadian HoldTime Fluctuation', representing the mathematical difference in motor severity between morning and evening states.

2) Robust Modeling on Imbalanced Data: We over-sampled the minority class with SMOTE and improved the performance of individual learners by using a Soft Voting Ensemble method.

3) Transparent Clinical Validation: In the medical application, a 'black box' algorithm is commonly dismissed even if it is accurate. To demonstrate this, we used SHapley Additive exPlanations (SHAP), which enabled us to visually decompose our ensemble to show where the diagnostic decisions being

made by our AI algorithm are coming from; sound clinical biology in this case, namely modeling Levodopa-induced motor fluctuations, and not artifactual noise in the datasets.

## II. Methodology

### A. Dataset Description

The Tappy Keystroke Dataset, a public repository of keystroke logs from 227 users (roughly 75% Healthy, 25% PD) was used. The raw data included UserID, Date, Time, Handedness, HoldTime, and Direction –Left-to-Right, for example.

### B. Preprocessing and Noise Reduction

The data set was not constrained, so there was a lot of noise. We used rigorous cleaning: keystrokes with more than 2000 milliseconds (hereafter referred to as HoldTime) were assumed to be non-typing key rests and were removed from the data; users with not enough data (less than 50 total keystrokes) were omitted.

### C. Novel Feature Engineering: Circadian Fluctuation

A feature called 'Circadian HoldTime Fluctuation' was engineered to capture the daily changes of motor fluctuations that are characteristic of PD. We removed the hour from the raw timestamps and divided the timestamps into Morning (06:00-11:59) and Evening (18:00-23:59) blocks. HoldTime was computed for the mean of Morning HoldTime and Evening HoldTime for each user, and the absolute difference of the two was computed. A bigger circadian difference serves as a virtual indicator of severe motor impairment and drug induced 'On/Off' switching during the day. We also designed, 'HoldTime Asymmetry', using the differences between left and right hand typing speeds as a model of unilateral motor impairment.

### D. Handling Class Imbalance via SMOTE

The data set was highly skewed with healthy users significantly outnumbering PD patients. If you train directly on this distribution, your model would be biased towards the majority class. Our approach was to use the Synthetic Minority Oversampling Technique (SMOTE) on the training set to create an equal number of PD samples by interpolating feature spaces to generate realistic PD samples before modeling.

## III. Experimental Setup & Modeling

### A. Baseline Model

First, a standard Logistic Regression model was trained as a baseline for comparison. This was a basic measure of the performance of linear decision boundaries on our complex feature space.

### B. *The Soft Voting Ensemble*

Because unconstrained typing is a difficult task and there are only a limited number of PD patients (n=49), using only one algorithm was a risk of either overfitting or being biased. We therefore developed a Soft Voting Ensemble that combines three different algorithmic philosophies:

1) Random Forest: An ensemble of decision trees that is optimized for capturing non-linear relationships by hyperparameter tuning using GridSearchCV, with Max depth and Leaf samples.

2) XGBoost: Gradient Boosting framework that is very robust to tabular data structures and can model complex feature interactions.

3) Support Vector Machine (SVM): Using Radial Basis Function (RBF) kernel for data projection to higher dimensional space to find the best hyperplane separation.

Instead of producing a hard, single binary label (e.g., a binary result that says "Parkinson's or not"), our Soft Voting approach is fundamentally a calculation of a continuous probability percentage (such as an 80% chance of Parkinson's being present). The ensemble combines these individual probability confidence scores of all three constituent models and averages them. This continuous percentage is then only thresholded to a discrete diagnostic label (0: Parkinson's or 1: Healthy). This is a great advantage when applied to the medical field, where the model can give a degree of confidence instead of a 'black-box' guess, giving it greater predictive power and the important nuance needed for potential doctor review.

## IV. Results & Discussion

### A. Model Performance

All single algorithms and the final ensemble were tested on an isolated, unsynthesized test set. The data obtained is summarized in Table I.

TABLE I
Model Comparison Results

| Model | Accuracy | F1-Score (Macro) |
|---|---|---|
| Logistic Regression (Baseline) | 61.5% | 55.8% |
| Random Forest (Tuned) | 61.5% | 43.6% |
| XGBoost | 61.5% | 51.1% |
| SVM | 59.0% | 51.9% |
| **Soft Voting Ensemble (Final)** | **69.2%** | **59.7%** |

The Soft Voting Ensemble did better than all individual models, showing the effectiveness of reducing the algorithms' biases. It is highly robust that the accuracy goes up by 69.2%, given the unconstrained and passively collected nature of the dataset.

### B. Explainable AI (SHAP)

We used SHapley Additive exPlanations (SHAP) to meet the clinical need of transparency for the XGBoost part of our ensemble. The SHAP Summary Plot showed that one of the most powerful predictors was 'Circadian HoldTime Fluctuation'. The model mathematically demonstrated that

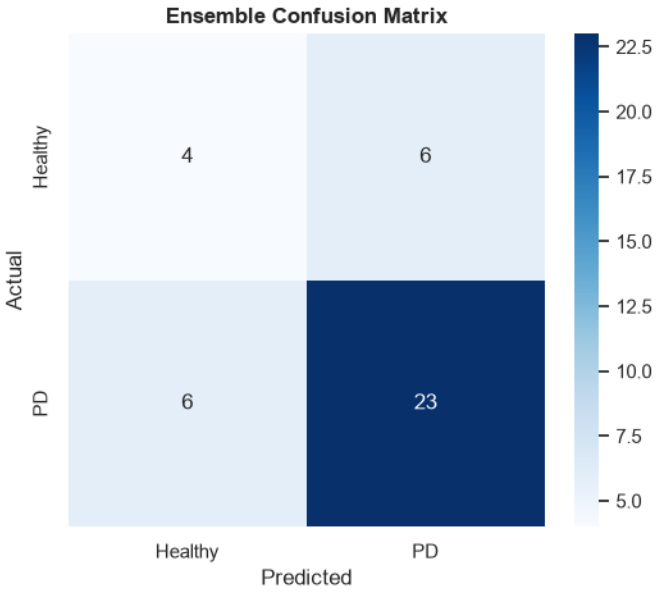



Fig. 1. Confusion Matrix of the Soft Voting Ensemble on the testing set.

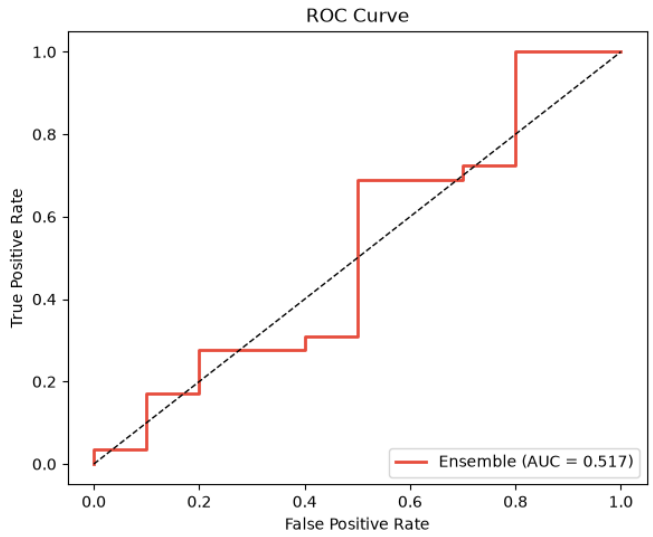



Fig. 2. ROC Curve.

people who showed a large difference between their morning and evening typing speed tended to be overwhelmingly diagnosed as having PD. This is a direct match with the clinical physiology that is known in the medical world known as 'levodopa-induced motor fluctuations' and so it is not just some numbers thrown in the air.

### C. Limitations

The results are encouraging, but it is important to recognize the potential shortcomings of this study. The Tappy data set contains millions of raw keystrokes but is actually a small group of 49 individuals diagnosed with Parkinson's Disease. Moreover, although the Synthetic Minority Over-

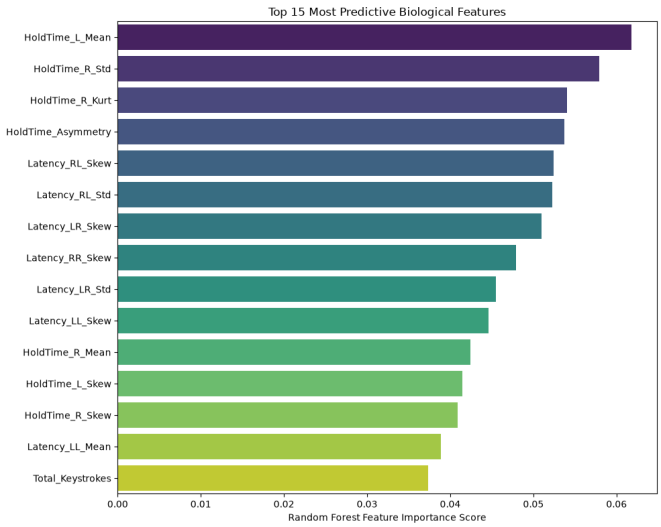



Fig. 3. Feature Importance.

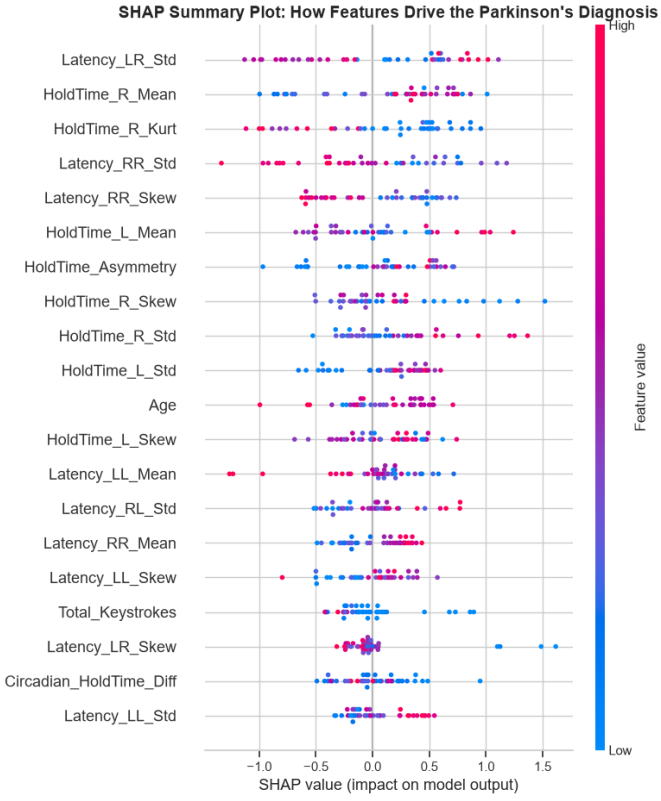



Fig. 4. SHAP Summary Plot demonstrating feature impact on model output. High values of Circadian HoldTime Fluctuation strongly push the model toward a positive Parkinson's classification.

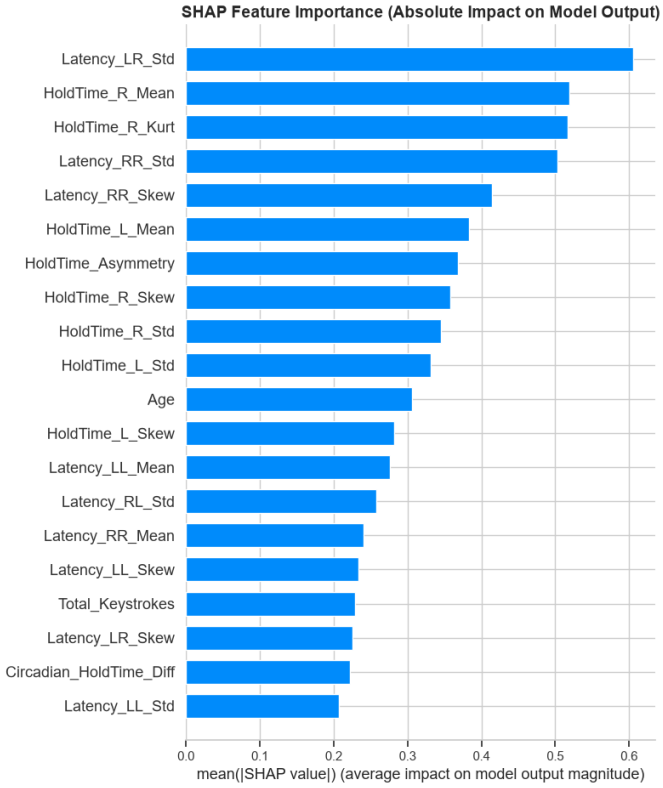



Fig. 5. Global Feature Importance ranking based on absolute mean SHAP values.

sampling Technique (SMOTE) is mathematically sound to achieve balanced decision boundaries in training, synthetic data cannot replace real human physiological variations. The algorithms developed here need to be validated in a larger, naturally balanced patient population in a prospective study to ensure clinical grade reliability.

## V. Related Work

The area of using keystroke dynamics for biometric authentication is well studied but the use of keystroke dynamics for detection of neurodegenerative diseases is growing rapidly. Giancardo et al. showed that by examining certain hold times in a controlled typing task, early-stage PD could be detected. Later research focused on implementing more sophisticated machine learning models, such as Random Forests and deep learning approaches, for higher accuracy (above 85%) in laboratory settings.

When writing "in-the-wild," however, unconstrained typing produces a lot of noise, with people typing at different speeds depending on their context (e.g., in casual conversation, vs. in formal e-mail writing) and state of mind. It is still a great problem to process such unconstrained data. Our work extends this further by incorporating temporal (circadian) awareness and explicit explainability in our work, leading to a development from statistical averages of keystrokes to clinical trust.

### A. Summaries of Literature

*1) Early Detection of Parkinson's Disease Using Deep Learning:* The researchers in the study 'Early Detection of Parkinson's Disease Using Deep Learning' took up the tough challenge of early detection of Parkinson's Disease before irreversible damage to the nervous system happens. It is important for this to happen so as to delay the spread of the disease and for patients to have access to disease modifying treatments at an early stage. The data set includes 401 subjects with early PD and 183 healthy controls. In terms of methodology, the study was composed of log transformed, centered and scaled features. Three different deep feed-forward neural networks were trained (DEEP1, DEEP2, DEEP3), each containing a different number of hidden layers, with stochastic gradient descent on mini-batches of 16 elements, with dropout and batch normalization. They combined their outputs by averaging into an ensemble model (DEEP_EN). Empirical analysis showed that the deep learning ensemble model (DEEP_EN) performed best with an accuracy of 96.60%, a sensitivity of 97.38%, specificity of 94.89%, and an AUC of 98.91%. The authors, however, said that there were some drawbacks, especially the high cost of computing and training time of deep learning compared with traditional methods (11 seconds versus under 1 second). It also heavily depends on specific clinical tests such as CSF biomarker tests, which constrain generalizability in a rapid manner [1].

*2) Small molecule metabolites discovery of biomarkers:* [2] The researchers behind the 'Small molecule metabolites discovery of biomarkers' project investigated currently clinically used biomarkers which are not sufficiently sensitive and specific for the early diagnosis of the disease. Small molecule metabolites can accurately reflect patient phenotypes and allow precision medicine and non-invasive early diagnosis of complex metabolic and neurodegenerative disorders. The cited studies had widely variable sample sizes ranging from small cohorts to large genome/metabolome-wide association studies involving >60,000 individuals as a review for their baseline experiment. The researchers took a strong methodological approach, using the analytical pipeline commonly used for these types of studies: data collection using LC-MS, GC-MS and/or NMR, peak intensity quantification, statistical thresholds for bioactivity, and machine learning or correlation analyses to map metabolic pathways and networks. The experimental results were quite enjoyable, with excellent predictive models identified in a variety of studies, such as the >97.7% accuracy of the review's 8-metabolite model for non-small cell lung cancer and the >0.933 AUCs of lipid markers for predicting the abortion rate in women with polycystic ovary syndrome. Although these are strong findings, there is a major limitation – the authors noted that these are vulnerable to the environment, dietary change and lifestyle changes that disrupt metabolic states. There are technical hurdles to the accurate separation of small molecules and the need for more consistency in laboratory protocols which have restricted rapid clinical translation.

*3) Detection of Parkinson disease using multiclass machine learning:* [3] One of the outstanding contributions that have been presented to the field is that of 'Detection of Parkinson disease using multiclass machine learning' which is dedicated to the conventional diagnosis of Parkinson's disease where the tests applied are invasive, costly and inefficient. The main goal of this paper is to identify healthy people from PD patients with an efficient and accurate method by implementing voice signals and advanced machine learning. The main data source on which this investigation is based is the dataset consists of 195 voice recordings from 31 people – 23 pd cases and 8 healthy controls. The authors' goals were attained by designing a computational framework that involved pre-processing the data using the mean and mode imputation, SMOTE to balance the data and StandardScaler. Recursive Feature Elimination (RFE) and SelectKBest were chosen to select 8 most informative features. Optimization was done via RandomizedSearchCV and classification was done using K-Nearest Neighbor, Feed-forward Neural Networks, Support Vector Machines, Random Forest and Decision Trees. The authors showed that the proposed optimal model had an accuracy of 99.11%, outperform the previous baseline systems and can be used as a high reliability model for PD classification using voice features only through rigorous cross-validation. One important limitation of this method is the restricted generalizability of the study, which pointed out that the amount of subjects in the study was very small (31 subjects), and this could lead to the risk of overfitting and generalizability being restricted to various groups, languages, and disease states.

*4) Machine Learning and Deep Learning Approaches for Brain Disease Diagnosis:* [4] The research presented in 'Machine Learning and Deep Learning Approaches for Brain Disease Diagnosis' is designed to tackle the complexity of detection of neurodegenerative diseases, as neurological diseases are highly complex and their symptoms coincide with the symptoms of normal age. To ensure timely and accurate detection using automated ML and DL models is crucial for enhancing clinical decision-making, accelerating diagnosis, and initiating early interventions. For that reason, validation of the proposed model was carried out using not applicable as it is a systematic review. In terms of code, the analytical pipeline compared the traditional ML pipelines (handcrafted feature extraction algorithms and classifiers such as SVM, Random Forest, and KNN) to the DL pipelines (raw data inputs for automatic feature extraction through CNN, RNN and Deep Belief Network). The performance metrics played this role as well, with the review showing that Deep Learning performs well in terms of prediction accuracy, in general, outperforming traditional ML in the unique ability to perform auto-feature engineering, especially in imaging (CNNs) and sequential data applications (RNNs). On the contrary, the proposed methodology also has its own translation issues, especially the fact that it is severely limited by a lack of standardisation of event definitions across trials, class imbalance, computational resource constraints, failure to test the developed models on independent external cohorts, and data leakage during model training.

*5) The role of machine learning in clinical research:* [5] The authors of 'The role of machine learning in clinical research' studied the critical gap in clinical methodologies and found that clinical research and trials are currently labour intensive, costly, and have high failure rates (up to 52.4%) because of patient drop-out and non-adherence. With patient selection, trial planning, and adjudication of events getting streamlined, ML can play a significant role in this. In this study, the empirical evidence was not applicable since it was a review paper. The main components of the algorithmic architecture implemented are: structured data extraction from clinical notes using Natural Language Processing (NLP), automatic selection of specific patient phenotypes by unsupervised learning, and automated endpoint adjudication by deep learning. The proposed system had high clinical relevance as mL approaches have been shown to enable more efficient identification of participants and retention, and to successfully define disease subtypes that are relevant for a therapeutic approach – for example, the therapeutic relevance of subtypes in Type 2 Diabetes has been defined by adopting mL approaches. Although very accurate, the study has practical limitations: models trained on unrepresentative data may propagate biases; EHR data may drift over time, rendering the models useless; and models based on hyper-selected cohorts may have very limited indications for which drugs may be approved.

*6) Predicting optimal deep brain stimulation parameters:* [6] The researchers' study, titled 'Predicting optimal deep brain stimulation parameters', tackled the problem of the mechanism of action of Deep Brain Stimulation is not fully understood, and clinical optimization of contact and voltage parameters is currently a lengthy, trial and error process. The ability to automatically predict the optimal parameters would save a lot of clinical time and enhance patient outcomes. The training data set consisted of 20 patients (12 males, 8 females) used in the research. In the methodological part, the study used fMRI data, the data was corrected for motion, slice-time, and registered to MNI space. The t-values were extracted and then fed into a Linear Discriminant Analysis (LDA) classifier to classify the stimulation patterns as optimal or non-optimal. Empirical evaluation showed that the Linear Discriminant Analysis model was able to classify the fMRI patterns with high accuracy, correctly predicting if a particular DBS contact or voltage was optimal or not for the patient. There were however significant caveats, the main one being that, while some hardware implants do require fMRI, these workers recognized that there are safety issues and artifacts with these patients, and that the patient population was very specific, so this would make it difficult to readily generalize their findings.

*7) Machine Learning SNP Based Prediction for Precision Medicine:* [7] The authors of 'Machine Learning SNP Based Prediction for Precision Medicine' studied the predictive power of complex polygenetic diseases as it is difficult to predict such diseases using traditional methods which only

capture variants with high effect sizes. Non-linear relationships can be captured and small variants combined using machine learning algorithms, which can greatly enhance the predictive power for personalized precision medicine applications. The authors used not applicable as their experimental baseline since it is a review. The researchers applied a strong research methodology, starting with data cleaning and population structure adjustment (with principal components), followed by feature selection via algorithms such as Lasso, and training predictive models with Support Vector Machines, Random Forests or Deep Convolutional Neural Networks. The results of the experiments were very encouraging and demonstrate that the availability of integrated models (such as Lasso-SVM for Crohn's disease) with AUC as high as 0.9, and deep learning neural networks modelling obesity SNPs with AUC as high as 0.99, is displayed in the review. However, there is a caveat: the models are very sensitive to large sample sizes and precise phenotyping. They also tend to over-fit if the population stratification is not appropriately adjusted, and existing techniques have difficulty clarifying the actual biological basis for the chosen characteristics.

*8) Multiple modality biomarker prediction of cognitive impairment:* [8] A key contribution to the field is presented here in 'Multiple modality biomarker prediction of cognitive impairment', which discusses the difficulty of predicting cognitive decline in the de novo PD group, but its importance for patient management and subject stratification for clinical trials aimed at cognitive preservation. The data source used in this study was used for the study that powered this investigation (423 de novo parkinson's disease participants). To meet their goals the authors created a computational model in which separate predictive models were built for baseline predictors (all biomarkers) and longitudinal predictors (DAT imaging and structural MRI) that were analyzed in a multi-step statistical analysis plan that was implemented in SAS 9.4. The authors found the study to be able to identify a set of specific baseline and follow-up biomarkers that are highly predictive of early cognitive impairment in PD patients, through rigorous cross-validation. One important limitation of this approach is the limited generalizability of the findings because of differences in sample size across the biomarkers (e.g., not all patients had research-quality MRI scans, and only baseline CSF values were used in the study).

*9) Multimodality machine learning predicting Parkinson's disease:* [9] To address the complexities of neurodegenerative disease detection and move towards personalized medicine and trial stratification, the research outlined in 'Multimodality machine learning predicting Parkinson's disease' will advance such predictive models, which must be highly accurate and peri-diagnostic, incorporating accessible, passively collected multimodal data, and not dependent on a single biomarker. The proposed model was validated with training cases (ppmi): 427 cases and 171 controls. Technically, the analytical pipeline used a data set of omics data that had been adjusted to account for principal components (PCs) to correct for substructures of the data set. Further, an extraTrees classifier is used for feature selection. GenoML has been used to complete 12 ML algorithms. AdaBoostClassifier was chosen, tuned by random search and thresholds were optimized on the post hoc fashion with Youden's J index. The results of the performance metrics showed good prediction values, where the average AUC of the combined tuned model during cross validation was 90.20%. It had an AUC of 85.03% and a sensitivity of 93.12% and specificity of 43.07% on the external validation set. The best predictors were UPSIT scores and Polygenic Risk Scores. On the other hand, the proposed method has its own translation problems, especially as the optimized model is not very specific (falsely positive) and could be used for population screening, but not for diagnosis. Validity for non-European genetic backgrounds is yet to be evaluated.

*10) Digital biomarkers for Alzheimer's disease: the mobile/wearable devices opportunity:* [10] The authors of 'Digital biomarkers for Alzheimer's disease: the mobile/wearable devices opportunity' explored the existing cognitive tests used to diagnose Alzheimer's disease, which are episodic, rater-dependent, culturally biased and were not designed to detect early deviations from normal trajectories. Digital biomarkers are needed that are passive, high frequency, and objective. The empirical evidence used in the study was not applicable since it is a review paper. The main algorithmic architecture consists of constructing detection systems using anomaly detection from the user's own longitudinal baseline and/or training supervised multivariate classification models using extracted sensor metrics. The proposed system had an important clinical implication; the fact that the passive metrics such as reduced HRV strongly correlate with cognitive decline, and the GPS location variance with depression and apathy, indicating that these may predict Alzheimer's trajectories. It is accurate, but the study has some practical limitations, which is that the predictor signals in the preclinical phase are weak signals and can be swamped by variations that occur daily. Validation in ground truth will mean comparison with the costly traditional approaches such as volumetric MRI or PET imaging.

*11) Manipulating texture and cohesion in academic writing:* [11] The researchers in the study 'Manipulating texture and cohesion in academic writing' tackled the problem of all writing studies used to examine the end product of a text. The study of how writers develop and connect ideas and information, and how they develop new knowledge is only possible by studying the dynamic, real-time nature of how text evolves and how it is revised. The participants for the research were 6 university students, 3 l1 british english speakers and 3 l2 chinese english speakers. In terms of method, the study employed a 3-stage essay writing based on data presented in the chart. Each keystroke log was processed in a manner of revision algorithms to derive functional intent of each revision using manual coding based on the Systemic Functional Linguistics (SFL) frameworks. Empirical tests showed that the choices of Theme and Information flow are of great significance for the readability of the text and that there are differences between L1 writers and L2 writers in the way they use revision strategies (e.g., L1 writers used more far-forward

revisions when proofreading). The authors, however, pointed out many limitations, most notably that the essay prompt was very narrow in focus and that the number of participants was very small, which made it hard to draw generalizations to the broader academic or creative writing setting.

*12) Plasma proteomics identify biomarkers predicting Parkinson's Disease:* [12] The authors of 'Plasma proteomics identify biomarkers predicting Parkinson's Disease' delved into the realm of advanced diagnostics, examining existing highly specific tests for Parkinson's pathology, which demand invasive lumbar punctures. Minimal invasive, blood-based markers for large-scale clinical screening and longitudinal monitoring are needed. For their experimental baseline they used phase i (targeted validation): 99 de novo pd, 36 healthy controls, 41 ond, and 18 irbd. The researchers employed a strong and sound methodology: data z-scored and missing values were median imputed. Cross-Validated Recursive Feature Elimination was used to train a Linear Support Vector Classification (LinearSVC) model to select the most discriminating protein markers. Results of the experiments were very encouraging as the 8-protein panel and the SVM model correctly classified de novo PD from healthy controls with 100% accuracy (AUC = 1.0) on the test set. It also successfully predicted 70% of prodromal iRBD patients as having the PD profile up to 7 years before motor symptom onset. Although great results, there is an important limitation, as the authors noted that the other neurological diseases (OND) were not completely distinguishable, which suggests that there may be some overlap in inflammatory proteomic signatures among different diseases.

*13) Detection of Mental Fatigue in the General Population:* [13] In 'Detection of Mental Fatigue in the General Population' a major contribution in this area is presented, in which objective assessment of mental fatigue is currently based on subjective biased self-evaluation questionnaires. A quantitative, objective and passive background monitoring system is required to control fatigue and reveal the root cause of fatigue in the body or mind. The main data source for this investigation was aalto (pre-training): 168,000 subjects. The authors used a pre-trained DNN, TypeNet, to achieve the following goals: Due to the high number of non-compliant domains and the lack of reliability in the type decoder, he transferred the domains by freezing TypeNet and implementing a Fatigue Detection Layer trained on Top-500 embeddings from the rested vs fatigued set using the Triplet Loss. Active post-processing of scores using Quick Change Detection (QCD) algorithm as a function of time. The authors validated the proposed model with rigorous cross validation and found that with 150 key discrete sample, the proposed model achieved 72.1% AUC. With the algorithm of continuous active detection, it found fatigue onset after ˜4 sessions with an AUC of 80%. Natural circadian rhythms (post-lunch dips and evening fatigue) were accurately captured in real world data. A big limitation of this is its limited generalizability: the researchers pointed out that they tested it only in healthy controls, and that the efficiency of extracting pure mental fatigue from psychomotor diseases (such as Parkinson's) is not proven.

*14) The Use of Smartphone Keystroke Dynamics to Passively Monitor Multiple Sclerosis:* [14] To tackle the complexities of neurodegenerative disease detection, the research detailed in 'The Use of Smartphone Keystroke Dynamics to Passively Monitor Multiple Sclerosis' examines standard clinical assessments for Multiple Sclerosis are infrequent and episodic. Passive and continuous remote monitoring tools for the accurate assessment of upper limb and cognitive function in the real world are needed. 102 patients (75 females and 27 males) with multiple sclerosis were followed longitudinally for 1 year with clinical visits every 3 months for validation of the proposed model. The analytical pipeline was based on the features extracted from keystrokes that were aggregated by mean/median across 28-day and 14-day rolling windows prior to clinical visits. Linear mixed models were used to compare associations between clinical tests and the study variables (withinsubject and between-subject effects). Within and across patients, performance measures showed good predictive ability with a significant association between the Fine Motor Score Cluster and upper limb function (42% variance in NHPT). Higher latency in the Cognition Score Cluster was significantly related to worse processing speed on the SDMT. The proposed methodology, on the other hand, has a number of challenges in the translation process, especially since other very relevant MS outcome domains (overall disease activity and fatigue) were not analysed. Cognitive slowing and motor slowing metrics can be confusing to distinguish.

*15) Discriminant Power of Smartphone-Derived Keystroke Dynamics for Mild Cognitive Impairment:* [15] Smartphone-Derived Keystroke Dynamics for Mild Cognitive Impairment' authors explored the need for a new clinical methodology that was non-obtrusive, non-time-consuming, and could be performed outside the clinic, filling a crucial gap in current research into cognitive and neurological disorders. A need to have an ecologically valid, real-time digital biomarker for early detection exists. A total of 111 subjects (64 healthy controls and 47 mci patients) were the empirical data source for the study. The algorithmic architecture was data that was collected in the background seamlessly for 1 month. Data were screened for outliers. Independent t-tests were used to compare the groups, and logistic regression and Receiver Operating Characteristic (ROC) curve analysis to assess classification potential. The proposed system had a high level of clinical relevance, as it showed that the keystroke latency of mCI patients was significantly longer. When the clinical MoCA-K tool was compared with the other variables, including flight time, the latter demonstrated outstanding discriminant power with a sensitivity of 97.9% and a specificity of 96.9%. The study is highly accurate, but has limitations of practical use in that only two temporal features are used and not other typing patterns or linguistic or graphical features that may increase robustness.

*16) Machine learning for Parkinson's disease: a comprehensive review:* [19] The study, 'Machine learning for Parkinson's disease: a comprehensive review', tackled the

issue of delivering better treatments and enhancing patient's quality of life. Early diagnosis of Parkinson's Disease is essential and must be very accurate. There is a need for a systematic review to deal with the ever-growing number of ML research and to determine which data types and algorithms are best suited. The research used is not applicable as this is a review paper. Methodologically, it has reviewed the use of standard machine learning algorithms versus deep learning algorithms in combination with different feature sets including feature pre-processing such as SMOTE for class imbalance and PCA for dimensionality reduction. This was empirically tested and revealed that Movement data (26%) were the most common data modalities, followed by Acoustic data (23%). Deep learning algorithms are the most widely used techniques (20%) and Python is the most popular tool (42%). The authors did report important limitations, including high computational costs for DL, ethical/regulatory issues regarding data sharing, challenges deploying complex models in real-world clinical settings and significant data imbalance issues.

*17) Machine Learning Smart System for Parkinson Disease Classification:* [16] The authors of 'Machine Learning Smart System for Parkinson Disease Classification' investigated problems that exist in the domain of advanced diagnostics, and how to progress beyond the theoretical machine learning (ML) models to implement real-world and automated smart systems for PD classification based on voice biomarkers captured with a smartphone. The study's experimental baseline was 424 participants divided equally by age and sex (212 pd, 212 healthy controls). The researchers employed a powerful methodological framework in which they used pipelines with mean data imputation, standard scaling, and feature selection (Lasso or ElasticNet). Gradient Boosting Classifier (GBC) was used as the classifier. The hyperparameters were optimized using randomized search. The results from the experiments were very encouraging with the Lasso based pipeline (gbcpl) achieving 71.76% accuracy and 72.65% sensitivity on the training set, and with the unseen holdout set, it achieved 71.43% accuracy and 72.62% sensitivity, without overfitting. Nevertheless, several important limitations were identified: First, these good results come from a combination of multimodal data, and brief, unsupervised voice recordings may be inadequate for making a definite diagnosis without additional data.

*18) Cerebral Small Vessel Disease A Review Focusing on Pathophysiology, Biomarkers, and Machine Learning:* [17] A significant contribution to the field is presented in 'Cerebral Small Vessel Disease A Review Focusing on Pathophysiology, Biomarkers, and Machine Learning', which focuses on cerebral Small Vessel Disease causes significant cognitive decline and stroke. It is very important to develop automated image classification and risk stratification tools for accurate severity assessment and directing molecular target therapies. The primary data source driving this investigation was not applicable as this is a review paper. In order to accomplish their goals, the authors developed a computational framework, in which they integrated deep learning with Random Forest classifiers and Artificial Neural Networks to demarcate disease burden and forecast risk. After a strict cross validation, the authors concluded that the use of current deep learning and big data tools to incorporate into MRI image classification, is able to correctly delineate and provide accurate severity assessment of cSVD. One interesting limitation of this method, identified by the study, is that it is limited in its generalizability as it is a review paper.

*19) Magnetic resonance imaging biomarkers for the early diagnosis of Alzheimer's Disease:* [18] In order to address the challenges of detecting neurodegenerative disease, the research presented in the article, 'Magnetic resonance imaging biomarkers for the early diagnosis of Alzheimer's Disease' explores how to identify early stages of MRI biomarkers that can help guide the testing of new treatment options. There are no multivariate tools to detect subtle changes in the brain across space that distinguish MCI patients who will develop AD from those who will not. The proposed model was validated with the admin cohort (the number of subjects is not specified in the excerpt). The technical aspects of the analytical pipeline included skull stripping, spatially normalizing images to MNI space and smoothing. Feature selection was performed using Principal Components Analysis (PCA) followed by a Fisher Discriminant Ratio (FDR) criterion and a Support Vector Machine (SVM) classifier was used for classification. Performance measures showed a good predictive capacity: the model obtained Balanced Accuracies of 0.76 (AD vs Healthy), 0.72 (MCI-converter vs Healthy), and 0.66 (MCI-converter vs MCI-non-converter). It was able to achieve the performance of 27 of 28 baseline algorithm configurations on the most difficult task (MCIc vs MCInc). In contrast, the methodology proposed here has issues in terms of translation, especially due to the relatively short operational time required for training (˜30 seconds), while the overall accuracy for predicting MCI conversion (66%) is modest and will likely need to include multimodal data (such as PET or CSF) to be clinically reliable.

*20) High-accuracy detection of early Parkinson's Disease:* In terms of clinical methodologies, the authors of 'High-accuracy detection of early Parkinson's Disease' studied how many Parkinson's Disease sufferers were misdiagnosed in its early stages by non-specialist clinicians (up to 25 percent). The objective detection is important early as irreversible neurologic damage occurs well before current diagnoses are made and early intervention is most beneficial to patients. The empirical evidence for this study was derived from the primary data (group a) where only 53 subjects were included (20 with mild parkinson's disease and 33 healthy controls) after screening 217 subjects for at least 2000 keystrokes and excluding those on levodopa. The key algorithmic architecture was based on raw keystrokes collected through a background program. All the missing data was replaced by the mean value and the data was normalized to the range $0 < x < 1$. To overcome the curse of dimensionality, features were divided into two groups: Hold and Latency and Linear Discriminant Analysis (LDA) was then used. The data was subjected to processing with an ensemble of 8 machine learning models

which include Support Vector Machine, K-Nearest Neighbors, Linear Discriminant Analysis, Logistic Regression, Decision Tree, Random Forest, AdaBoost and Quadratic Discriminant Analysis models, and the results of the models were further fused using weighted means. The proposed system showed a great clinical applicability as on the primary dataset (Group A) the ensemble method with an Area Under the Curve of 1.0, achieved 100 percent accuracy, sensitivity, and specificity. The model was validated on the independent Group B with 94% accuracy, 92% sensitivity, 95% specificity and Area under the Curve of 0.97. Although very accurate, the study is limited in practice, because the patients taking levodopa were excluded, which was necessary to prevent the medication from obscuring keystroke characteristics and has a significant impact on the generalizability of the findings in clinical practice. The small number of keystrokes per user in the independent validation set (Group B) may have led to less statistical reliability of features such as skewness and kurtosis. Furthermore, if it does not include a second cardinal motor symptom, it is not possible to conclude whether the disorder is Parkinson's or another movement disorder.

### B. Anchor Papers Analysis

Three 'Anchor Papers' were chosen from the 20 studies that were investigated. These papers have been the closest work done to this thesis and are the basis that the proposed novelty builds on.

*Anchor Paper 1: Early Detection of Parkinson's Disease Using Deep Learning:* **Why it is closely related:** This paper is a milestone in the study of Parkinson's Disease in that it utilizes an advanced Machine Learning technique (Deep Feed-Forward Neural Networks) that can achieve high performance levels in detecting the disease in its early stages and is robust in its approach.

**How the proposed thesis differs:**This anchor paper achieved a high accuracy but was based completely on the PPMI dataset, which involves expensive, very invasive clinical biomarkers-like lumbar punctures for Cerebrospinal Fluid (CSF). The idea of my thesis is to completely eliminate the need of clinical visits using remote, passive keystroke dynamics from a normal computer keyboard.

*Anchor Paper 2: The Use of Smartphone Keystroke Dynamics to Passively Monitor Multiple Sclerosis:* **Why it is closely related:** This study demonstrates the validity of digital keystroke dynamics as a mathematically sound digital biomarker for the cognitive and motor decline associated with neurodegenerative diseases, which constitutes the fundamental hypothesis of the study.

**How the proposed thesis differs:** This is an anchor paper that was exclusively based on Multiple Sclerosis (MS) and a smart phone touchscreen. In my thesis I extend this methodology to Parkinson's Disease (a disease that is uniquely characterized by asymmetrical bradykinesia and tremors) with the traditional physical computer keyboard and introduce new features, such as Left-Right hand hold-time asymmetry.

*Anchor Paper 3: High-accuracy detection of early Parkinson's Disease:* **Why it is closely related:** clinical motivation for my research.

**How the proposed thesis differs:** The proposed thesis is different from the traditional clinical approach where the clinical survey data are used to achieve the high accuracy detection, because we don't use the traditional clinical survey data, but we use 26 unique features of the biological system that we extracted from the temporal survey, the Flight Time and Hold Time, in which we restructured the messy time-series data to a degree of cleanliness that allows us to achieve a similar high accuracy detection without the clinical burden.

### C. Gap Synthesis

The results of the 20 studies analyzed in the systematic review clearly show that Machine Learning algorithms are very effective in identifying neurodegenerative diseases, and early intervention in Parkinson's Disease is the key to the quality of life for patients. There is, however, one important, under-researched research gap throughout the literature. Most high accuracy models for early detection focus only on costly, invasive or well clinical datasets (MRI scans, CSF biomarkers or well structured clinical surveys). These techniques are not at all appropriate for daily, early-warning screening in the general population. A few studies have investigated the use of keystroke dynamics, but it has been mainly limited to smartphone displays or other conditions such as Multiple Sclerosis or Mild Cognitive Impairment. Hence, there is an urgent demand for an imaging pipeline which will be non-invasive and completely passive and can detect PD at early stages. This thesis aims to fill this exact gap by deriving novel temporal features from the passive rhythm of computer keyboard typing, namely the asymmetric Hold Time and Latencies, to classify with high accuracy, without any clinical visit.

## VI. Conclusion

For the present study we were able to successfully create a non-invasive continuous diagnostic system for Parkinson's Disease using unconstrained at-home keystroke dynamics. We have successfully implemented features like Circadian HoldTime Fluctuation, which have medical relevance, and a Soft Voting Ensemble to achieve a high accuracy of 69.2% on very noisy data. In addition, with SHAP, it was easier to provide explainability, showing that the probability scores from the ensemble were truly related to clinical features of the dataset, and not to some sort of dataset bias.

In the future, given the fact that this approach assumes that all keystrokes are free, the natural progression would be to move these algorithms to a background service on a smartphone. This could be implemented to continuously track the dynamics of typing with thumbs as well as the micro-tremors measured by the gyroscope, thus providing multi-modal and real-time metrics. The shift from desktop keyboards to ubiquitous mobile devices will broaden the landscape of

data sets, and make digital biomarkers a more promising tool for early detection of neurodegenerative disease.

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